

Heard but Not Heeded

Paralinguistic Information Encoding and Loss in Audio-Language Models

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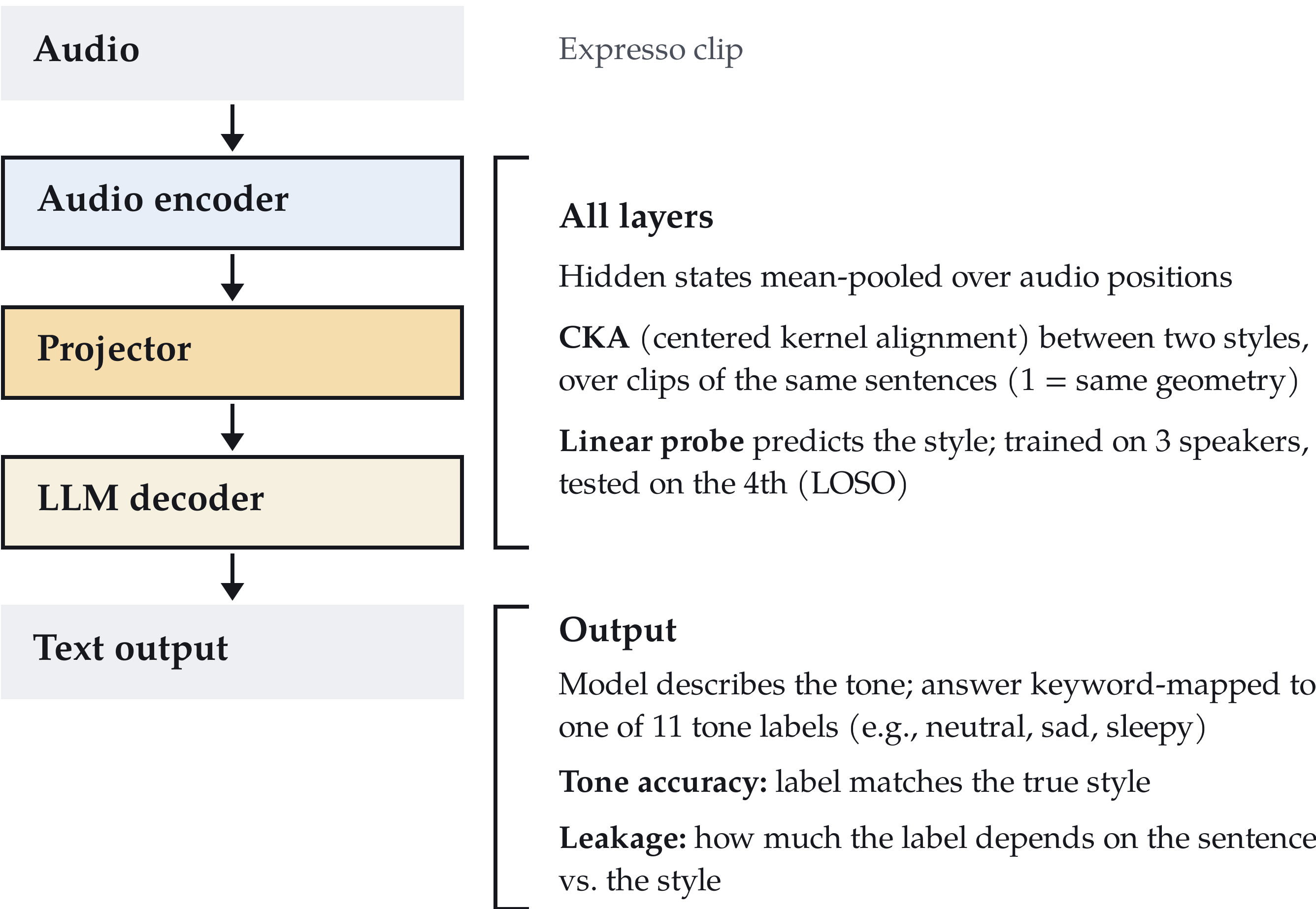
1 Motivation

Audio-language models (ALMs) sometimes appear insensitive to tone. In three open ALMs, plus Whisper as a reference, we ask whether paralinguistic information (how something is said; here, speaking style) is absent from their internal representations, or present but not reflected in their answers.

2 Setup

Espresso (Nguyen et al., 2023): 4 speakers read the same sentences in 7 styles (confused, default, enunciated, happy, laughing, sad, whisper), 10,636 clips in total. We pair clips with the same sentence and speaker but different styles, giving 31,896 pairs.

3 Methods



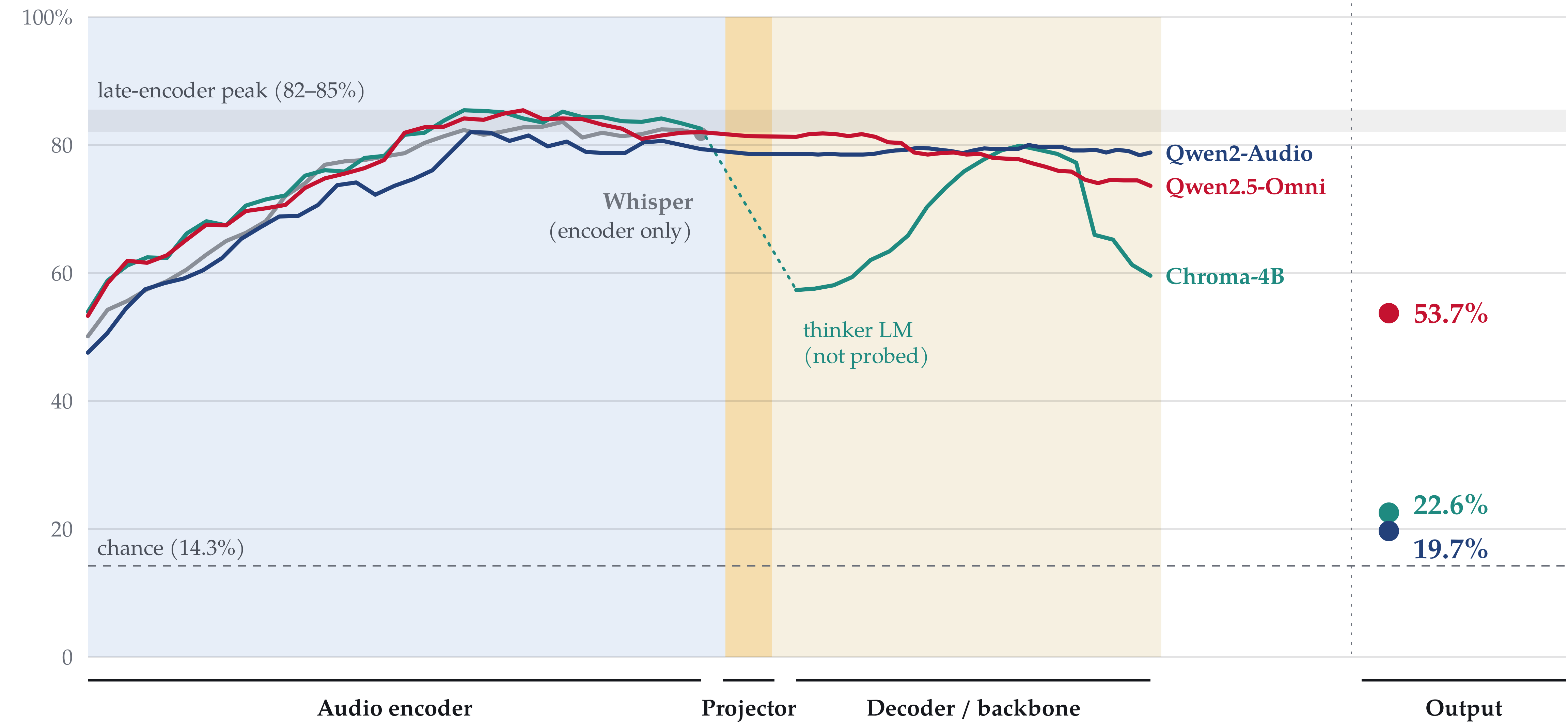
Models

Model	Layers probed	Objective
■ Whisper-large-v2	32 encoder	Speech recognition
■ Qwen2-Audio-7B-Instruct	33 encoder → linear projector → 33 LLM	Text generation
■ Qwen2.5-Omni-7B	32 encoder → MLP projector → 28 LLM	Text + speech generation
■ Chroma-4B	32 audio encoder → 16 speech backbone → 4 speech decoder	Text (thinker) + speech with voice cloning

The projector maps audio-encoder outputs into the LLM’s input space. Whisper gives no tone answer. Chroma’s language model (“thinker”, 36 layers) sits between its audio encoder and speech backbone and is not probed. Its CKA across styles is 0.98–1.0.

4 Encoder–output gap

Lines: linear probe accuracy (7-way, LOSO) at each layer, with stages aligned across models. **Dots:** tone accuracy of the model’s text output. The probe is trained and the tone answer is zero-shot, so the two are not directly comparable.

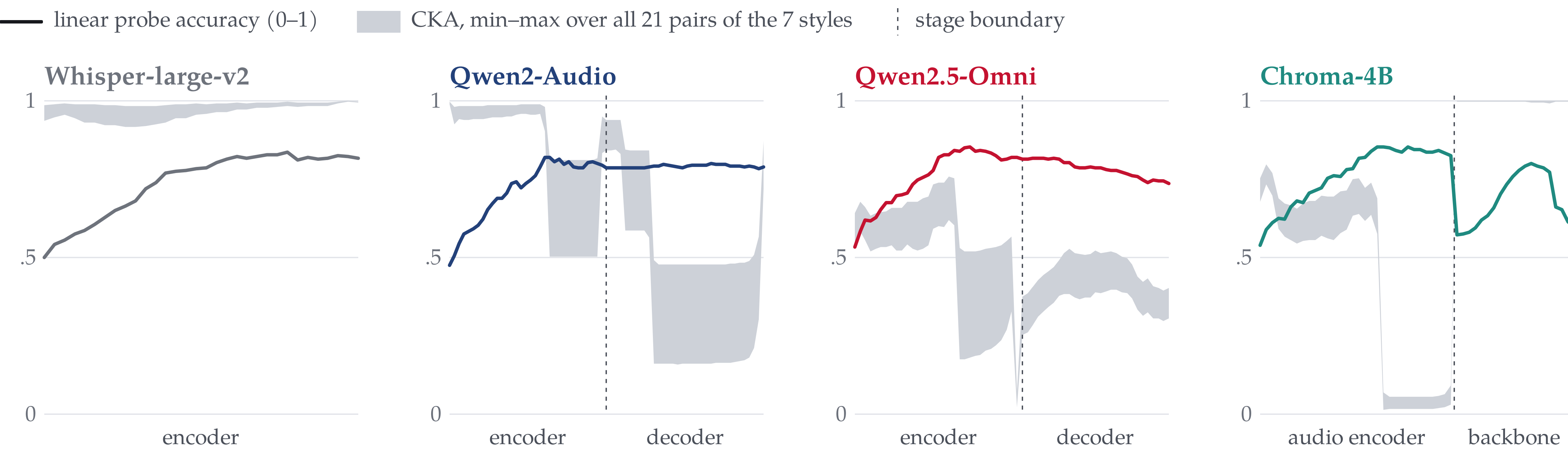


Late encoder
82–85% probe accuracy for all four models.

Projector
78–81%, while CKA changes sharply.

Decoder / backbone
Tone accuracy 19.7% (Qwen2-Audio), 22.6% (Chroma), 53.7% (Qwen2.5-Omni).
81% → 74%; Chroma 58% at backbone entry, ~80% late in the backbone, ~60% at exit.

CKA vs. linear probe



CKA stays within 0.92–1.0 while probe accuracy rises from ~50% to over 80%, so high CKA does not mean style information is absent.

CKA shifts at the projector and drops in the decoder, while probe accuracy stays near 79%.

Probe accuracy declines through the decoder, from 81% (projector) to 74% (last layer).

In the backbone, CKA $\approx$ 1.0 (all styles share one geometry), yet probe accuracy recovers to ~80%.

Takeaways

- Speaking style is linearly decodable.** A linear probe recovers the style (7 classes, e.g., happy, whisper) at 82–85% from the late audio encoder of all four models (chance 14.3%).
- Asked to describe the tone, the three audio-language models are right only 19.7–53.7% of the time.** Qwen2-Audio’s and Chroma’s answers follow the sentence more than the style (leakage 0.98 and 0.84, where 1 = fully sentence-driven). Qwen2.5-Omni’s follow the style more (0.27).
- Why Omni differs is not settled.** Our leading hypothesis is its speech-generation objective, but it also uses an MLP projector rather than a linear one. Retraining only Qwen2-Audio’s linear projector with gradient reversal did not change leakage (0.977 → 0.979).

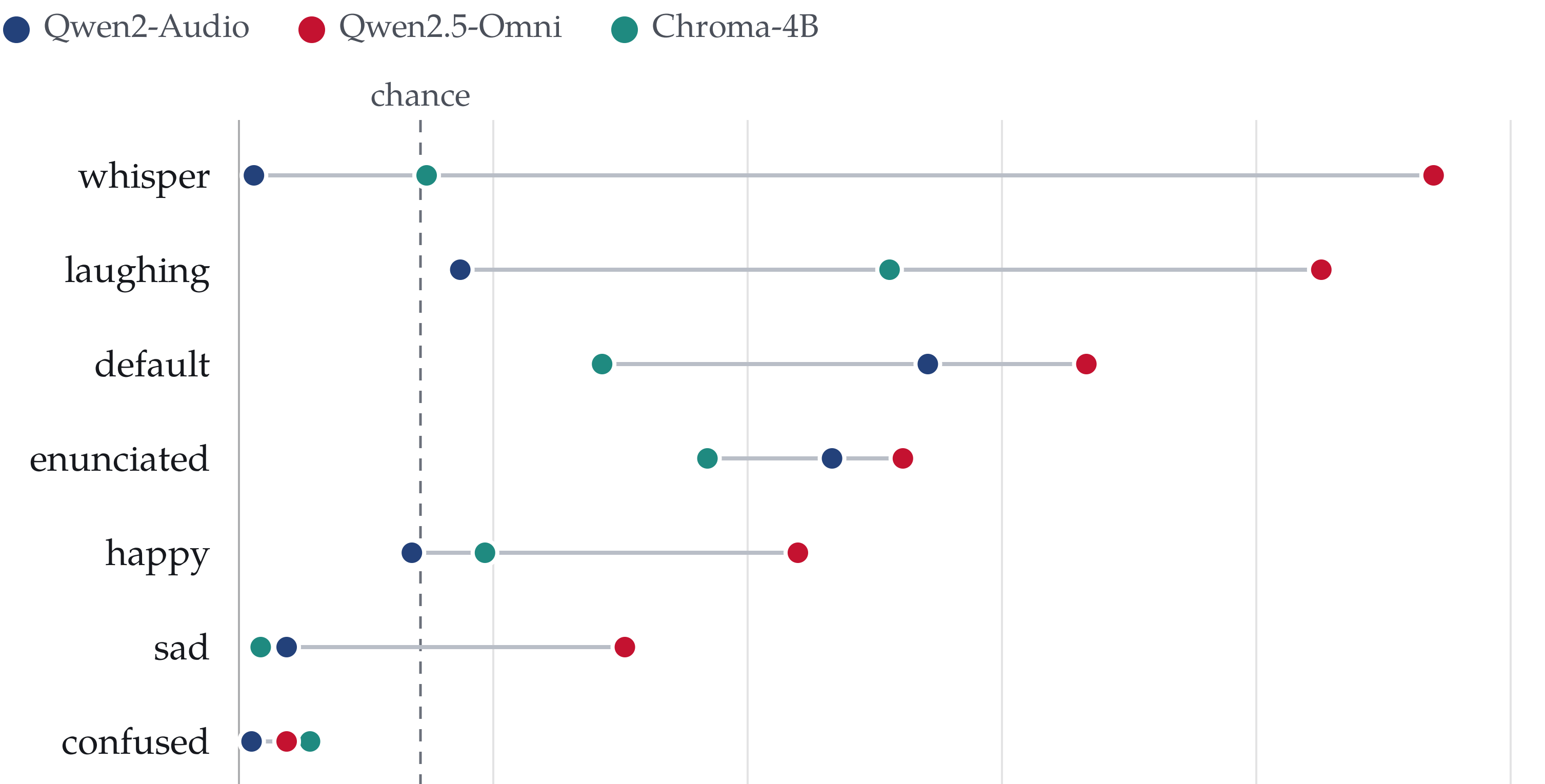
5 Output-level behavior

Prompt: “Reply in English only. What is the person saying? Describe the tone briefly (e.g. neutral, happy, sad, angry, whispering, confused, laughing).” Leakage uses normalized mutual information (NMI) over all clips to compare how much the tone label depends on which sentence was spoken vs. how it was spoken.

Model	NMI(label, sentence)	NMI(label, style)	Leakage	Same label in all 7 styles
■ Qwen2-Audio	0.234	0.006	0.977	30.5%
■ Qwen2.5-Omni	0.140	0.374	0.272	0.6%
■ Chroma-4B	0.254	0.047	0.844	19.9%

Leakage is near 1 when the label tracks the sentence and near 0 when it tracks the style. The last column counts sentences whose label is the same in every style.

Per-style tone accuracy



Styles are sorted by Qwen2.5-Omni’s accuracy. Gray lines join the three models in each row.

Example outputs

“Monday there’s gonna be haze but Tuesday look for thunderstorms”				
Model	sad	whisper	laughing	confused
■ Qwen2-Audio	neutral	neutral	neutral	neutral
■ Qwen2.5-Omni	sleepy	whispering	happy	neutral
■ Chroma-4B	neutral	neutral	neutral	neutral

Tone label each model gives for the same sentence in four spoken styles. Omni labels sad speech “sleepy”, which we count as incorrect. It may reflect a mismatch in tone vocabulary.

6 Discussion

- Qwen2-Audio and Qwen2.5-Omni have similar encoders but leakage of 0.98 vs. 0.27. Omni is also trained for speech generation and uses an MLP projector. With two models, we cannot separate these factors.
- Chroma’s backbone receives inputs with CKA $\approx$ 1.0 across styles, yet its probe accuracy recovers to ~80%. This suggests the backbone partly re-encodes style from context.

Retraining the projector

We retrain only Qwen2-Audio’s linear projector (1280-dim → 4096-dim, ~5M params) with a style-classification head and an adversarial content head that predicts the sentence (20 text clusters) through a gradient reversal layer (GRL), to keep style decodable while suppressing content.

	Baseline	GRL
Leakage ratio	0.977	0.979
NMI(label, style)	0.006	0.004
NMI(label, sentence)	0.234	0.203
Tone accuracy	19.7%	18.0%

The style head reached 93.3% validation accuracy by epoch 4. The checkpoint we evaluated (epoch 2, weakest content head) left leakage unchanged.

Limitations

- One dataset (Espresso) with only 4 speakers, so results may not generalize to larger speaker populations
- One prompt, keyword-mapped to tone labels
- Mean pooling removes temporal dynamics (e.g., laughing)
- Probe accuracy shows that style is decodable, which does not mean the model uses it



Paper
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